Lecture 6: Confidence Intervals

IEOR E4709: Statistic Analysis and Time Series
©2025 by Agostino Capponi

1 Properties of MLE Estimator:

Let $\hat{\theta}_{MLE}$ be the maximum likelihood estimator (MLE) of θ. The following properties hold:

- **Consistency.** $\hat{\theta}_{MLE}$ converges to θ in probability.
- **Asymptotic Normality.** $I(\theta)^{1/2}\left(\hat{\theta}_{MLE}-\theta\right)$ converges in distribution to a standard Gaussian vector, or equivalently, $\hat{\theta}_{MLE}$ follows a Gaussian distribution with mean θ and covariance matrix $I(\theta)^{-1}$. (A matrix B is said to be the square root of A if $BB=A$). Here, $I(\theta)$ is the Fisher information matrix computed using the entire set of observations $X_1, X_2, \ldots, X_n$.
- **Asymptotic Efficiency.** An estimator is said to be efficient if its variance equals the Cramer-Rao lower bound $I(\theta)^{-1}$. Since the MLE estimator $\hat{\theta}_{MLE}$ achieves this lower bound asymptotically (i.e., as the number of observations goes to infinity), it is said asymptotically efficient.
- **Invariance.** If $\hat{\theta}_{MLE}$ is the MLE of θ and g is a function, then $g(\hat{\theta}_{MLE})$ is the MLE of $g(\theta)$. For the proof, see the book "Statistical Inference" by Casella and Berger.

2 Interval Estimation

We have so far discussed about point estimation. In this section, we discuss *interval estimation*, i.e., a range of values computed from data which is predicted to contain the population parameter with a specified probability.

An *interval estimator* of a parameter θ is a pair of estimators $L(X), U(X)$, such that $L(X) \leq U(X)$. Consequently, $[L(X), U(X)]$ is an interval. We are interested in the coverage probability of an interval estimator, i.e., in computing $\mathbb{P}_\theta\left(\theta \in [L(X), U(X)]\right)$, where $\mathbb{P}_\theta$ indicates that the samples $X=(X_1,\ldots,X_n)$ are being generated from the probability distribution parameterized by the true parameter θ.

The *confidence coefficient* is the infimum, over all probabilities, that a confidence interval will contain the true value of the population parameter θ wit the desired confidence (i.e.. with the specified probability) For example, if the confidence coefficient is 0.95, then it means that $\inf_{\theta\in\Theta}\mathbb{P}_\theta\left(\theta \in [L(X), U(X)]\right) = 0.95$, where Θ is the space of allowed parameters.

An interval estimator with a specific confidence coefficient is called a *confidence interval*.

Example: Consider i.i.d. samples $X_1, X_2, \ldots, X_n$ from a Gaussian distribution with *unknown mean* μ and known variance σ^2. Our objective is to construct an interval estimator of μ with a high confidence coefficient. We know that the sample mean $\bar{X}$ has a Gaussian distribution with mean μ and variance σ^2/n. Therefore, $\frac{\bar{X}-\mu}{\sigma/\sqrt{n}}$ follows a Gaussian distribution with mean 0 and variance 1. Then we have that

$$\begin{aligned}\mathbb{P}_\mu\left(\frac{|\bar{X}-\mu|}{\sigma/\sqrt{n}} \leq c\right) &= \mathbb{P}_\mu\left(\bar{X} - c\sigma/\sqrt{n} \leq \mu \leq \bar{X} + c\sigma/\sqrt{n}\right) \\ &= 1 - 2\Phi(-c),\end{aligned}$$

where $\Phi(x)$ is the cumulative distribution function of a standard (zero mean unit variance) Gaussian random variable. Notice that the interval is independent of the unknown parameter μ. Let $z_{1-\alpha}$ be defined as the quantile of the distribution with tail probability α, which means that $\mathbb{P}(Z > z_{1-\alpha}) = \alpha$. In order to have $1-\alpha$ confidence coefficient (for instance, $\alpha = 5\%$), we want $1 - 2\Phi(-c) = 1 - \alpha$, i.e. $\Phi(-c) = \frac{\alpha}{2}$. This means that $-c = z_{\frac{\alpha}{2}}$, and, because the Gaussian distribution is symmetric, this implies $c = -z_{\frac{\alpha}{2}} = z_{1-\frac{\alpha}{2}}$. The method described above yields exact confidence intervals, where the probability of coverage is at least the claimed $(1-\alpha)$.

Remark: In the previous example we have constructed an exact confidence interval. Many times, this is not possible, and one settles for confidence intervals that are calculated by means of an approximation (for example, a normal approximation). Suppose that $\hat{\theta}_{MLE}$ is the MLE of a scalar parameter θ. Because of asymptotic normality, we know that $I(\theta)^{1/2}\left(\hat{\theta}_{MLE} - \theta\right)$ follows approximately a standard Gaussian random variable. An approximate confidence interval of θ with coefficient $1-\alpha$ is then $\left[\hat{\theta}_{MLE} + z_{\frac{\alpha}{2}} I(\theta)^{-1/2}, \hat{\theta}_{MLE} + z_{1-\frac{\alpha}{2}} I(\theta)^{-1/2}\right]$. Such an approximate confidence interval is computable if $I(\theta)$ is known.

Remark 2.1. *If we were dealing with a vector $\theta = (\theta_1, \ldots, \theta_d)$, then the confidence interval of θ with coefficient $1-\alpha$ would be a hyper-rectangle, and given by $\left[\hat{\theta}_{MLE} + z_{\frac{\alpha}{2}} diag(I(\theta)^{-1/2}), \hat{\theta}_{MLE} + z_{1-\frac{\alpha}{2}} diag(I(\theta)^{-1/2})\right]$. Here, $diag(A) = (A_{11}, A_{22}, \ldots, A_{dd})^\top$ is the column vector consisting of the diagonal entries of the matrix A. Again, such an approximate confidence interval is computable if $I(\theta)$ is known.*

2.1 Estimating $I(\theta)$

In the case of a Gaussian random variable, we know that the sample mean is the MLE of the population mean μ. We also know that the MLE is asymptotically efficient and hence $I(\mu) = \frac{n}{\sigma^2}$. Hence, it is computable and independent of the parameter μ which we are trying to estimate. In many practical circumstances, however, the Fisher information of a parameter θ, $I(\theta)$, is not known, or difficult to compute. Indeed, if $I(\theta)$ depends on θ, we cannot compute it because we do not know θ (otherwise, we would not be trying to estimate it). In this case, it is important to estimate $I(\theta)$. There are three approaches which can be used:

- **Method 1.** If we can compute $I(\theta)$ explicitly, then we know by the invariance property that $I(\hat{\theta}_{MLE})$ is a maximum likelihood estimator of $I(\theta)$. Moreover, by the properties of the maximum likelihood estimator, we know that $I(\hat{\theta}_{MLE})$ is asymptotically unbiased. We then just use $I(\hat{\theta}_{MLE})$ to construct the confidence interval.

- **Method 2.** If we cannot compute $I(\theta)$ explicitly, then we can plug $\hat{\theta}_{MLE}$ inside the defining expression and estimate $I(\theta)$ as $V_E(\hat{\theta}_{MLE})$, where $V_E(\theta) = -\mathbb{E}_\theta \left[\frac{\partial^2 \log L(\theta|X)}{\partial\theta\partial\theta^\top} \right]$. We can approximate the latter quantity numerically using Monte-Carlo methods, i.e., $V_E(\hat{\theta}_{MLE}) = -\sum_{i=1}^{n} \frac{1}{n} \frac{\partial^2 \log L(\theta|X_i)}{\partial\theta\partial\theta^\top}\Big|_{\theta=\hat{\theta}}$, where X_i is the realization of the data vector X in the i-th Monte-Carlo iteration.

- **Method 3** If we cannot compute $I(\theta)$ explicitly, we can use a sample mean estimation, given by $V_B(\hat{\theta}_{MLE})$, where

$$V_B(\theta) := \sum_{i=1}^{n} \frac{1}{n} \left(\frac{\partial \log L(\theta|X_i)}{\partial\theta} \right) \left(\frac{\partial \log L(\theta|X_i)}{\partial\theta} \right)^\top,$$

  where X_i is the realization of the data vector X in the i-th Monte-Carlo iteration.